MANIANANTH M

Software Developer | Python Backend & Flutter Applications

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PROFESSIONAL SUMMARY

Software Developer with 3 years of combined experience across software development and manufacturing, focused on Python backend development and cross-platform Flutter applications. Skilled in FastAPI REST APIs (with working knowledge of Django and Flask), PostgreSQL, MySQL, Redis, JWT/RBAC security, and real-time systems using WebSockets and MQTT. Experienced with Docker, Linux, Nginx, and CI/CD; familiar with Celery, Pytest, and microservices architecture.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, Flask, REST API, Async, SQLAlchemy, Pydantic, Celery, Pytest, RAG (basic knowledge)

Databases: PostgreSQL, MySQL, Redis

Architecture & Fundamentals: Microservices, MVVM, System Design, Data Structures & Algorithms (DSA)

Security & Communication: JWT, OAuth 2.0, RBAC, RSA, WebSockets, MQTT

DevOps & Cloud: Docker, Docker Compose, Linux, Nginx, Git, GitHub Actions, CI/CD, AWS, Hostinger VPS

Frontend & Tools: Flutter, Dart, Provider, GoRouter, HTML, CSS, Postman, SQL

PROFESSIONAL EXPERIENCE

SPI Equipments India Pvt. Ltd. — Software Developer, R&D | 2024 – Present | Coimbatore, Tamil Nadu

- Develop and maintain cross-platform business applications using Flutter and Dart.
- Build REST APIs and backend services using Python and FastAPI with PostgreSQL and MySQL databases.
- Implement JWT authentication and role-based access control; build real-time workflows using WebSockets and asynchronous programming.
- Containerize backend services with Docker and Docker Compose, and configure Nginx as a reverse proxy.
- Use Git, GitHub, and GitHub Actions to manage source code and support CI/CD workflows.

SELECTED PROJECTS

Service Software — After-Sales & Service Management Application

Technology: Flutter, Dart, Provider, GoRouter, FastAPI, Python, MySQL, WebSockets, Firebase Cloud Messaging, Docker, Nginx

- Developed a cross-platform service-management application supporting Admin, Service Engineer, Customer, and Accounts workflows.
- Built FastAPI endpoints with JWT authentication and role-based access control; implemented real-time updates using WebSockets and Firebase Cloud Messaging.
- Supported cloud deployment with Docker containers and Nginx reverse-proxy routing.

ColorChef V4 — Industrial Dye Dispensing SCADA

Technology: Flutter Desktop, Dart, MVVM, RS-232 Serial, Multithreading, SQLite, Hive

- Developed an industrial desktop HMI/SCADA application for automated textile dye and chemical dispensing, controlling multi-axis (X/Y/Z) mechanical positions and 24-beaker batch scheduling.
- Integrated continuous RS-232 industrial weighing scale telemetry using background Dart Isolates for zero-latency dosing precision, storing batch history in an offline SQLite database with PDF/Excel reporting.

Yarn Quality Control & Lea Strength Tester

Technology: Flutter Desktop, Dart, MVVM, RS-232 Serial, SQLite FFI, fl_chart, PDF Generation

- Developed an industrial textile QC desktop application to measure yarn tensile breaking strength, automatically calculating Yarn Count (Ne), CSP, Standard Deviation, and CV%.
- Integrated dual RS-232 serial interfaces for digital precision scales and machine load controllers, generating real-time analytics graphs and certified laboratory PDF test reports.

Restaurant Self-Ordering Kiosk & Kitchen Display System

Technology: Python, FastAPI, PostgreSQL, Redis, WebSockets, Docker Compose, Nginx

- Designed a restaurant self-ordering workflow connecting customer kiosks with a kitchen display system.
- Designed FastAPI backend services and PostgreSQL data structures for menu, order, and kitchen workflows, with WebSocket-based real-time order updates and Redis in the architecture.
- Prepared a containerized deployment using Docker Compose and Nginx.

EDUCATION & CERTIFICATION

B.E. — Mechanical Engineering — Adhiyamaan College | 2015 – 2019 | CGPA: 8.01

Python Full Stack Developer Certification — PUMO TECHNOVATION | 2023